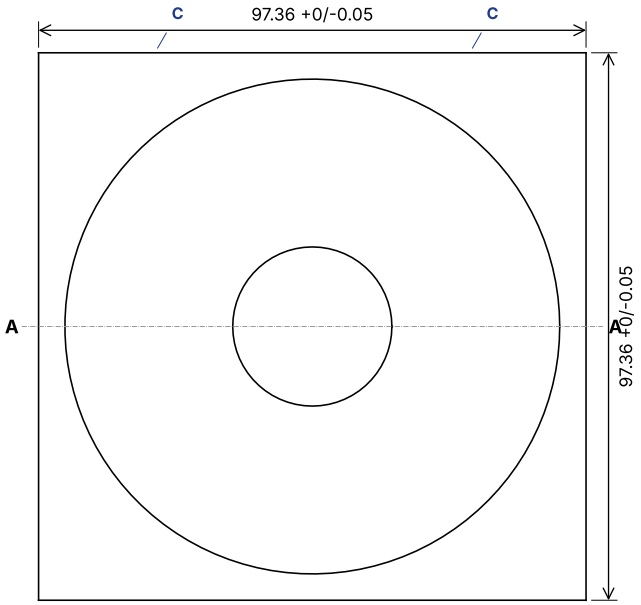
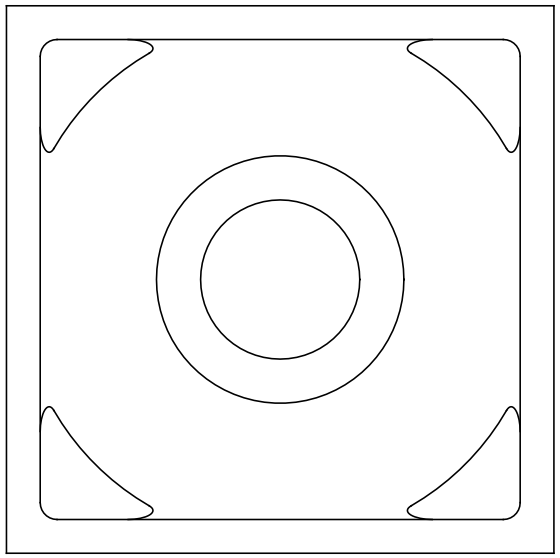
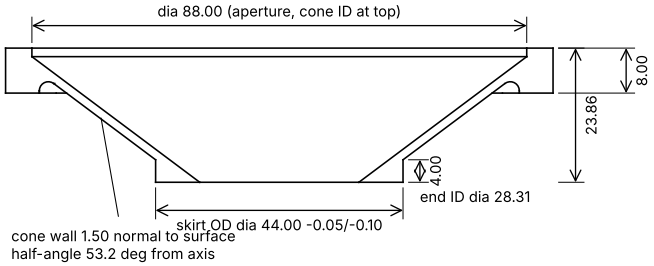
TOP VIEW (outer face = aperture plane)



BOTTOM VIEW



underside pocket: square 85.36, R3.0,
to 1.52 below the outer face



SECTION A-A (through the axis, seen from -Y)

FEATURE TABLE

A Aperture and cone ID: dia 88.00 +-0.10 at the plane 1.52 below the outer face, straight through the 1.52 web; the cone ID narrows at 53.19 deg half-angle to dia 28.31 at the skirt end face. Concentric with outline within 0.1.

W Cone wall 1.50 +-0.10 measured normal to the surface (OD is the ID offset 2.50 radially).

K Skirt at the small end: cylindrical OD dia 44.00 -0.05/-0.10, concentric with the cone ID within 0.05, running 4.00 up from the end face to where it meets the cone OD. End face flat and square to the axis within 0.05, 23.86 +-0.05 below the outer face. The skirt runs inside the H09 collar with 0.5 radial clearance and must not touch it.

P Underside pocket around the cone: square 85.36 with R3.0 corners, 6.48 deep from the inner rim face, leaving a 6.0 rim and a 1.52 web.

C 8x M2.5×0.45 INSERT into the four edges, 2 per edge at +-28.0 from centre, 4.76 below the outer face, thru the rim into the pocket

F Outer face flat within 0.05 (aperture plane). CONE ID FINISH: after turning, bead blast or etch to a matte texture of 0.1 rad rms facet slope or more; turning marks alone do not satisfy this. Black anodize or low-outgassing black coating follows for flight; as-blasted is acceptable for this unit.

M Suggested process: mill square, pocket corners and edge holes; turn cone ID from the aperture side, and the cone OD, skirt OD and end face from below, in a 4-jaw chuck. 3D-contoured cone OD also acceptable.

R Inside corners: R1.6 fillet along the four junctions of the cone OD with the flat pocket walls, blended into the web. Other pocket corners R1.5 max (R3.0 vertical corners as drawn). Where the skirt OD meets the cone OD: tool nose R1.0 max.

NOTES

Units: millimetres. Third-angle projection. Tolerances unless stated: features and hole positions +-0.1; outline +-0.2.
Threads: ISO metric coarse, 6H. Holes marked INSERT are tapped for stainless free-running helical coil inserts, length 1.5D, supplied and installed by the shop; drill and tap to the insert maker's data. Inserts flush to 0.25 below.
Every drilled and tapped hole runs through to an open volume (venting). Break sharp edges 0.1-0.3; deburr all holes.
Finish: as machined for this unit. Please also quote: clear anodize MIL-A-8625 Type II Class 1, with the H01 host face and the panel/angle faces under rivets masked and conversion coated MIL-DTL-5541 Type II Class 3 (electrical bonding).
Assembly is by Carthage: fasteners locked and torqued per the SLANT assembly procedure (to be issued).
Fit-check unit built to the flight layout. Material certs appreciated; no inspection report required.
Model: the STEP file of the same part number governs any dimension not shown here.
The cone is optical: no dents or tool marks on its inside surface. Protect it in packing.

PART	**SLANT-H02 Top plate with integral baffle**
MATERIAL	6061-T6 plate, 1 in stock (finished 23.86)
QTY	1
MASS (model)	84 g
PROJECT	SLANT 1U housing, assembly rev 16
OWNER	Carthage College, Space Sciences
DRAWN / CHECKED	from model 2026-10-08 / _____
SHEET REV	D